Pharmacology of local anesthetics

3 BDS Lecture

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Course Learning Outcomes

- List the different local anesthetic agents used in dentistry
- Define the pharmacological properties of local anesthetic agents
- Compare different local anesthetic agents

Pharmacokinetics of Local Anesthetics

When injected into soft tissues, local anesthetics produce vasodilation

Ester local anesthetics are also potent vasodilating drugs.

Procaine, the most potent vasodilator among local anesthetics, is injected clinically to induce vasodilation when peripheral blood flow has been compromised because of (accidental) intra-arterial (IA) injection of a drug (e.g., thiopental) or injection of epinephrine or norepinephrine into a fingertip or toe.

Cocaine is the only local anesthetic that consistently produces vasoconstriction.

Issues with Vasodilation

- A significant clinical effect of vasodilation is an increase in the rate of absorption of the local anesthetic into the blood,
- thus decreasing the duration and quality (e.g., depth) of pain control,
- Increasing the anesthetic blood (or plasma) concentration and its potential for overdose (toxic reaction).

Oral Route

- With the exception of cocaine, local anesthetics are absorbed poorly, from the gastrointestinal tract after oral administration.
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- In addition, most local anesthetics (especially lidocaine) undergo a significant hepatic first-pass effect after oral administration

Topical Route

- Local anesthetics are absorbed at differing rates after application to mucous membrane:
- In the tracheal mucosa, absorption is almost as rapid as with intravenous (IV) administration (intratracheal drug administration [epinephrine, lidocaine, atropine, naloxone, and flumazenil] is used in certain emergency situations);
- in the pharyngeal mucosa, absorption is slower

- Applied to intact skin, they do not provide an anesthetic action, but with skin damaged by sunburn, they bring rapid relief of pain.

- Sunburn remedies usually contain lidocaine, benzocaine, or other anesthetics in an ointment formulation.

- EMLA (eutectic mixture of anesthetics)
- EMLA cream (lidocaine 2.5% and prilocaine 2.5%)
- Is an oil emulsion
- Able to provide surface anesthesia for intact skin.
- Because intact skin is barrier to drug diffusion , EMLA cream to be applied 1 hour before procedure.
- Extraction of mobile primary teeth roots, pulpal therapy is possible

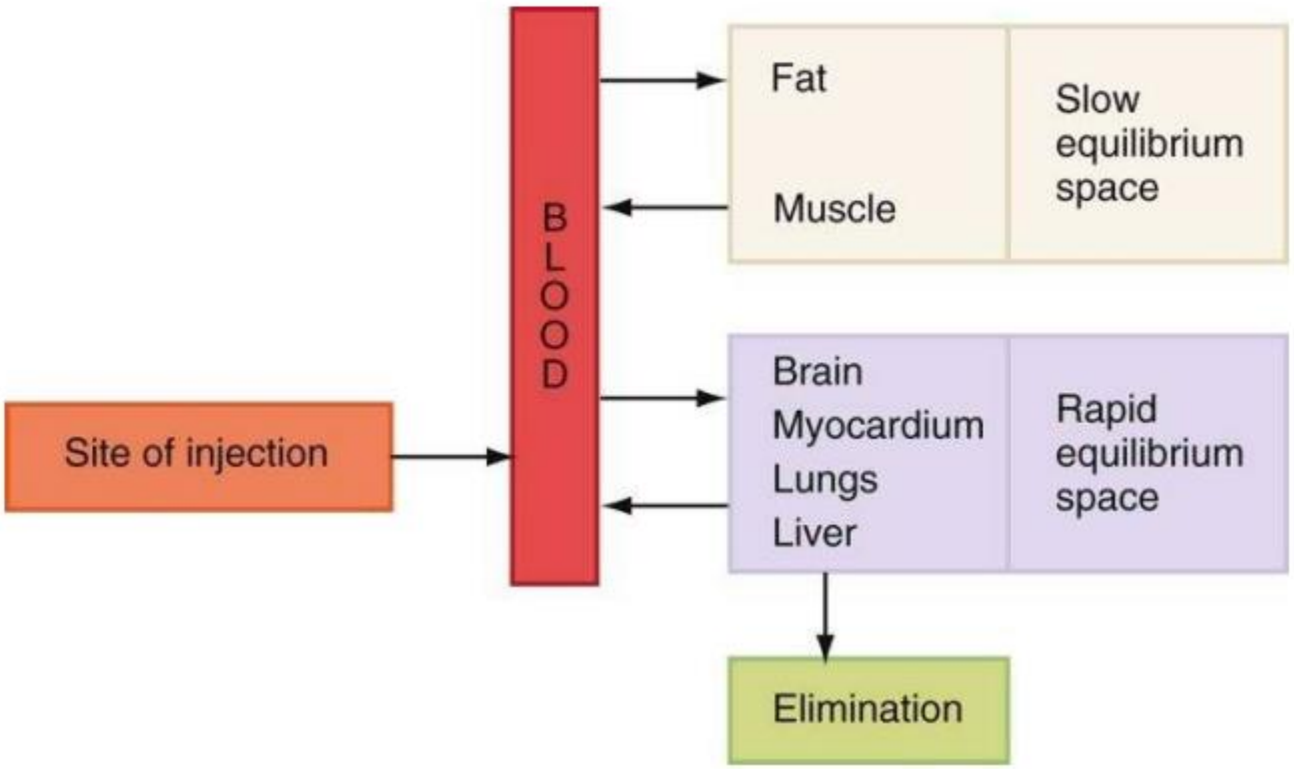
Injection

- The absorption rate of local anesthetics after parenteral administration (subcutaneous, intramuscular, or IV) is related to both the vascularity of the injection site and the vasoactivity of the drug.
- IV administration of local anesthetics provides the most rapid elevation of blood levels and is used clinically in the primary management of ventricular dysrhythmias.

Distribution

- Once absorbed into the blood, local anesthetics are distributed throughout the body to all tissues
- Highly perfused organs , such as the brain, head, liver, kidneys, lungs, and spleen, initially will have higher anesthetic blood levels than less perfused organs.
- Skeletal muscle contains the greatest percentage of local anesthetic of the body.

Pattern of distribution of local anesthetics after absorption



Half-Life of Local Anesthetics

The rate at which a local anesthetic is removed from the blood is described as its elimination half-life.
The elimination half-life is the time necessary for a 50% reduction in the blood level

Drug	Half-life, hours
Chloroprocaine*	0.1
Procaine*	0.1
Tetracaine*	0.3
Articaine†	0.5
Cocaine*	0.7
Prilocaine†	1.6
Lidocaine†	1.6
Mepivacaine†	1.9
Ropivacaine†	1.9
Etidocaine†	2.6
Bupivacaine†	3.5
Propoxycaine*	NA

NA, Not available.

* Ester.

† Amide.

Ester Local Anesthetics

- Ester local anesthetics are hydrolyzed in the plasma by the enzyme **pseudocholinesterase.**
- Chloroprocaine, the most rapidly hydrolyzed, is the least toxic, whereas tetracaine, hydrolyzed 16 times more slowly than chloroprocaine, has the greatest potential toxicity.
- Procaine undergoes hydrolysis to **para-aminobenzoic acid** (PABA), which is excreted unchanged in the urine
- Allergic reactions that occur in response to ester local anesthetics usually are not related to the parent compound (e.g., procaine) but rather to **PABA**, which is a major metabolic product of many ester local anesthetic.

Contraindications to Ester drugs

- Approximately 1 of every 2800 persons has an atypical form of pseudocholinesterase, which causes an inability to hydrolyze ester local anesthetics and other chemically related drugs (e.g., succinylcholine).
- Its presence leads to prolongation of higher local anesthetic blood levels and increased potential for toxicity.
- Atypical pseudocholinesterase is a hereditary trait. Any familial history of difficulty during general anesthesia should be carefully evaluated by the doctor before
- dental care commences.

Amide Local Anesthetics

- The primary site of biotransformation of amide local anesthetics is the liver.
- Prilocaine undergoes primary metabolism in the liver, with some also possibly occurring in the lung.
- Articaine, a hybrid molecule containing both ester and amide components, undergoes metabolism in both the blood and the liver.

- Patients with lower than usual hepatic blood flow (hypotension, congestive heart failure) or poor liver function (cirrhosis) are unable to biotransform amide local anesthetics at a normal rate.
- This slower than normal biotransformation rate results in higher anesthetic blood levels and increased risk of toxicity
- This represents a relative contraindication to the administration of amide local anesthetic drugs

Lidocaine Disposition in Various Groups of Patients

Group	Lidocaine Half-Life, hr
Normal	1.8
Heart failure	1.9
Hepatic disease	4.9
Renal disease	1.3

Excretion

- The kidneys are the primary excretory organ for both the local anesthetic and its metabolites.
- Esters appear only in very small concentrations in the urine because they are hydrolyzed almost completely in the plasma.
- Amides usually are present in the urine in a greater percentage than the esters, primarily because of their more complex process of biotransformation.
- Patients with significant renal impairment may be unable to eliminate the local anesthetic compound or its major metabolites from the blood, resulting in slightly elevated blood levels and therefore increased potential for toxicity.

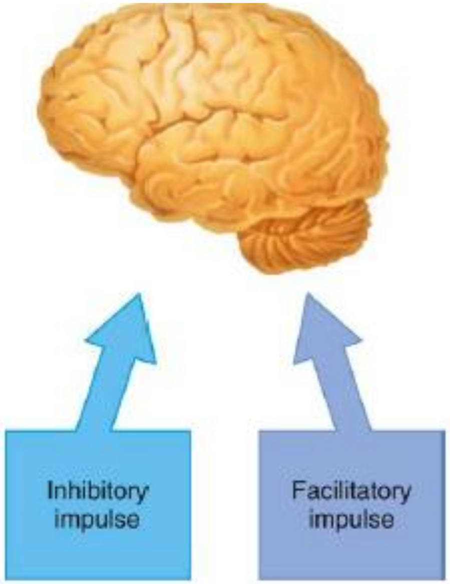
Systemic Actions of Local Anesthetics

- Local anesthetics are chemicals that reversibly block action potentials in all excitable membranes.
- The central nervous system (CNS) and the cardiovascular system (CVS) therefore are especially susceptible to their actions.

Central Nervous System

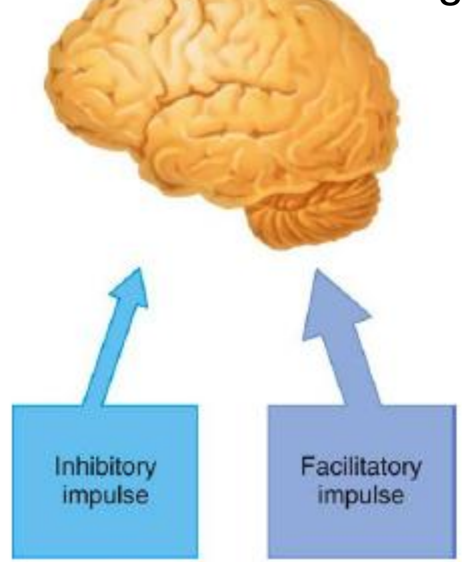
- Local anesthetics readily cross the blood–brain barrier. Their pharmacologic action on the CNS is seen as depression.
- At low (therapeutic, nontoxic) blood levels, no CNS effects of any clinical significance have been noted.
- At higher (toxic, overdose) levels, the primary clinical manifestation is a generalized tonic– clonic convulsion.
- Between these two extremes exists a spectrum of other clinical signs and symptoms

Local Anesthetic action on inhibitory and facilitatory impulses in a cerebral cortex

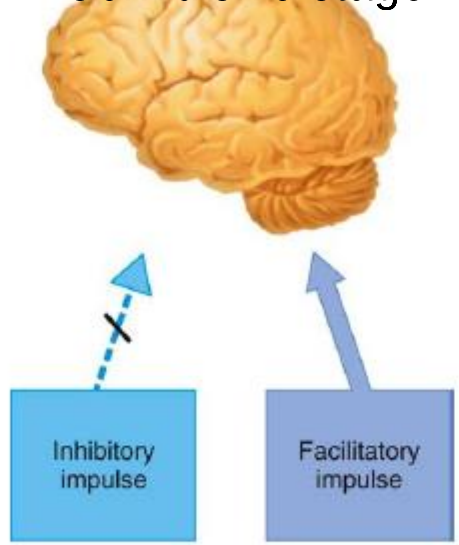


Normal cerebral cortex

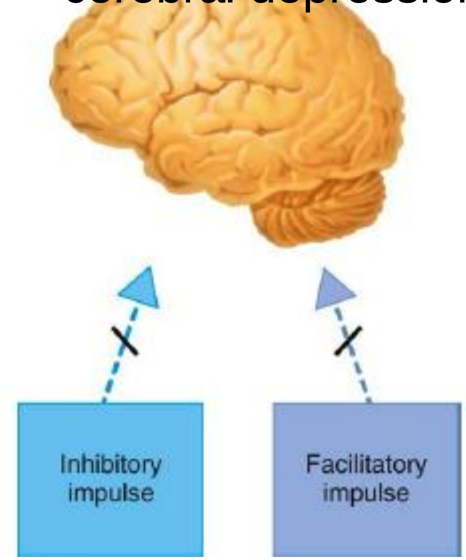
Pre convulsive stage



Convulsive stage



Final stage – cerebral depression



Anticonvulsant Properties

- Some local anesthetics (e.g., procaine, lidocaine, mepivacaine, prilocaine, even cocaine) have demonstrated anticonvulsant properties

Clinical Situation	Lidocaine Blood Level, µg/mL
Anticonvulsive level	0.5-4
Preseizure signs and symptoms	4.5-7
Tonic–clonic seizure	>7.5

Preconvulsive Signs and Symptoms of Central Nervous System Toxicity

Signs (objectively observable)	Symptoms (subjectively felt)
Slurred speech	Numbness of tongue and circumoral region
Shivering	Warm, flushed feeling of skin
Muscular twitching	Pleasant dreamlike state
Tremor of muscles of face and distal extremities	
Generalized lightheadedness	
Dizziness	
Visual disturbances (inability to focus)	
Auditory disturbance (tinnitus)	
Drowsiness	
Disorientation	

Direct Action on the Myocardium

- Local anesthetics modify electro conduction in the myocardium
- As the local anesthetic blood level increases, the rate of rise of various phases of myocardial depolarization is reduced. So Local anesthetics produce a myocardial depression
- Local anesthetics decrease the electrical excitability of the myocardium, decrease the conduction rate, and decrease the force of contraction.

- Therapeutic advantage is taken of this depressant action in managing the hyperexcitable myocardium, which manifests as various cardiac dysrhythmias

- At blood levels greater than the therapeutic (anti-dysrhythmic) level include a decrease in myocardial contractility and decreased cardiac output, both of which lead to circulatory collapse.

Minimal to Moderate LA Overdose Levels

Signs	Symptoms (progressive with increasing blood levels)
Talkativeness	Lightheadedness and dizziness
Apprehension	Restlessness
Excitability	Nervousness
Slurred speech	Sensation of twitching before actual twitching is observed (see "Generalized stutter" under "SIGNS")
Generalized stutter, leading to muscular twitching and tremor in the face and distal extremities	Metallic taste
Euphoria	Visual disturbances (inability to focus)
Dysarthria	Auditory disturbances (tinnitus)
Nystagmus	Drowsiness and disorientation
Sweating	Loss of consciousness
Vomiting	
Failure to follow commands or be reasoned with	
Elevated blood pressure	
Elevated heart rate	
Elevated respiratory rate	

Moderate to High Overdose Levels

- Tonic–clonic seizure activity followed by

- Generalized central nervous system depression
- Depressed blood pressure, heart rate, and respiratory rate
- Death

Contraindications for Local Anesthetics

Medical Problem	Drugs to Avoid	Type of Contraindication	Alternative Drug
Local anesthetic allergy, documented	All local anesthetics in same chemical class (e.g., esters)	Absolute	Local anesthetics in different chemical class (e.g., amides)
Bisulfite allergy	Vasoconstrictor-containing local anesthetics	Absolute	Any local anesthetic without vasoconstrictor
Atypical plasma cholinesterase	Esters	Relative	Amides
Methemoglobinemia, idiopathic or congenital	Prilocaine	Relative	Other amides or esters
Significant liver dysfunction (ASA 3–4)	Amides	Relative	Amides or esters, but judiciously
Significant renal dysfunction (ASA 3–4)	Amides or esters	Relative	Amides or esters, but judiciously
Significant cardiovascular disease (ASA 3–4)	High concentrations of vasoconstrictors (as in racemic epinephrine gingival retraction cords)	Relative	Local anesthetics with epinephrine concentration of 1 : 200,000 or 1 : 100,000, or mepivacaine 3%, or prilocaine 4% (nerve blocks)
Clinical hyperthyroidism (ASA 3–4)	High concentrations of vasoconstrictors (as in racemic epinephrine gingival retraction cords)	Relative	Local anesthetics with epinephrine concentration of 1 : 200,000 or 1 : 100,000, or mepivacaine 3%, or prilocaine 4% (nerve blocks)

Local Anaesthesia Agents

A. ESTERs Group

1. Procaine(Novacain)

- **Chemistry:**

-Ester

-Metabolism :hydrolyzed rapidly in plasma by pseudocholinesterase

- **Pharmacology:**

- Weak anaesthetic with low toxicity

- Vasodilator so the use of VasoConstrictor agent is a must

-Used in a 2% concentration which have a sufficient potency and low toxicity

-Slow onset -takes 6 to 10 minutes

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- Procaine is of importance in the immediate management of inadvertent intra-arterial (IA) injection of a drug;
- its vasodilating properties are used to aid in breaking arteriospasm.
- The incidence of allergy to both procaine and other ester local
- anesthetics is significantly greater than to amide local anesthetics.
- The maximum recommended dose of procaine, used for peripheral nerve blocks, is 1000 mg.
- With a dissociation constant (pKa) of 9.1, procaine has a slow clinical onset of anesthesia (6 to 10 minutes)

CASE REPORT

Necrosis of the thumb after inadvertent injection of diclofenac in the radial artery: a case report

Abstract | **Fulltext** | Metrics | Get Permission | Cite this article

Authors Shon HC, Park JK, Kang SW, Yang JY

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Abstract: Inadvertent intra-arterial drug injection occurs

B. Amide Group

1. Lidocaine Hcl

- **Chemistry:**

  - Amide ,effective dental concentration 2%

pKa 7.9.

pH of Plain Solution 6.5.

pH of Vasoconstrictor-Containing Solution

≈3.5.

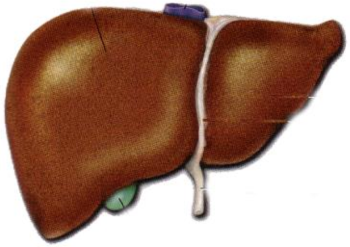
- Lidocaine is the Gold standard for comparison

-onset of action-Rapid (3 to 5 minutes)

- **Biotransformation**

  -Unlike esters group, amide group undergo biotransformation mainly in the liver rather than hydrolyzed in plasma.

  -So duration of anesthesia is longer



- **Maximum recommended dose:**
- The maximum dose recommended by FDA of lidocaine with or without vasoconstrictor is **7.0mg/kg of bodyweight** for adult and pediatric patient, not to exceed an **absolute maximum dose of 500 mg.**
- **(one cartridge of 2% contains 36 mg)**

	duration of anesthesia	
	pulpal	soft tissue
Lidocaine HCl 2% with epinephrine 1:100000	60 min	180-300 min
Lidocaine 2% without vasoconstrictor	5 – 10 min	

Figure 4-2 A, Lidocaine 2%. B, Lidocaine 2% with epinephrine 1:50,000. C, Lidocaine with epinephrine 1:100,000.

- 2.Mepivacaine Hcl (3%)
- Amide
- Potency: 1 (Lidocaine = 1).
- Slight vasodilatation
- Duration of pulpal anesthesia: Mepivacaine without a vasoconstrictor is 20 to 40 minutes. 2 to 3 hours of soft tissue anesthesia
- **Onset –** Rapid (3 to 5 minutes).
- <u>Effective Dental Concentration</u>
- 3% without a vasoconstrictor; 2% with a vasoconstrictor.
- Maximum recommended dose -6.6 mg /kg not to exceed 400 mg
- (one cartridge of 3% contains 30*1.8= 54 mg)

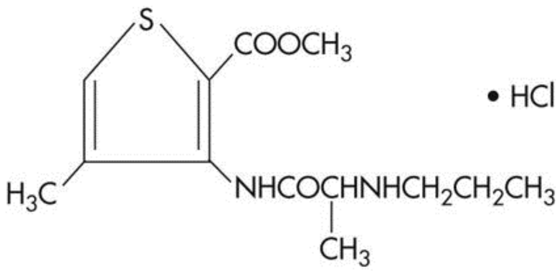
Figure 4-3 **A** through **C,** Mepivacaine 3%. **D,** Mepivacaine 2% with levonordefrin 1:20,000.

3.Articaine Hcl (4%)

- Hybrid molecule- classified as an amide
- But possesses both amide and ester character.
- Potency 1.5 (lidocaine =1)

- Biodegradation occurs in both plasma and liver
- Onset of action: infiltration 1 to 2 min
- mandibular block 2 to 3 min

- maximum recommended dose 7.0 mg/kg
- effective concentration – 4%
- (one cartridge contains 40*1.8= 72mg)



- **duration of anesthesia**

	pulpal	soft tissue
Articaine HCl 4% with epinephrine 1:100000	**60- 75 min**	**180-360 min**

- Articaine is able to diffuse through the hard and soft tissues better than most local anesthetics.
- Clinically, after maxillary buccal infiltration Articaine may provide palatal soft tissue anesthesia, so palatal injections are not needed

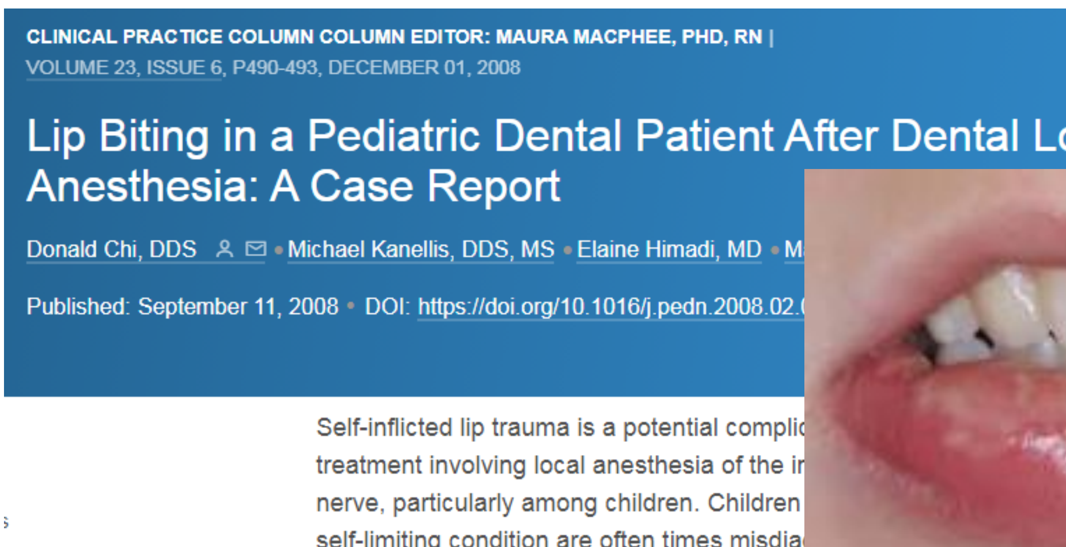
Bupivacaine HCl(0.5%)

- Amide
- Potency:Four times that of lidocaine, mepivacaine, and prilocaine.
- Onset of Action :Slower onset time than other commonly used local anesthetics (e.g., 6 to 10 minutes).
- Effective Dental Concentration :0.5%.
- Anesthetic Duration : pulpal 90-180 minutes Soft tissue 4 to 8 hours
- Maximum Recommended Dose
- The FDA maximum recommended dose of bupivacaine is 90 mg

indications

- 1 Lengthy dental procedures for which pulpal (deep) anesthesia in excess of 90 minutes is necessary
- (e.g., full mouth reconstruction, implant surgery, extensive periodontal procedures)
- 2 Management of postoperative pain (e.g., endodontic, periodontal, postimplant, surgical)
- The patient's requirement for postoperative analgesics is considerably lessened when bupivacaine
- is administered for pain control.
- For postoperative pain control after a short surgical procedure (<30 minutes), bupivacaine may be administered at the start of the procedure

- Bupivacaine is not recommended in younger patients or in those for whom the risk of postoperative soft tissue injury produced by self-mutilation is increased, such as physically and mentally disabled persons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Lip Biting in a Pediatric Dental Patient After Dental Lo
Anesthesia: A Case Report

Donald Chi, DDS • Michael Kanellis, DDS, MS • Elaine Himadi, MD • M

Published: September 11, 2008 • DOI: https://doi.org/10.1016/j.pedn.2008.02.

Self-inflicted lip trauma is a potential complic
treatment involving local anesthesia of the in
nerve, particularly among children. Children
self-limiting condition are often times misdia

Topical Anaesthetic Agents

Topical anaesthesia is obtained by direct application of the drug to the mucous membrane or the skin

The concentration of a local anesthetic applied topically is greater than that of the same local anesthetic administered by injection

.

The higher concentration facilitates diffusion of the drug through the mucous membrane.

Higher concentration also increases the risk of toxicity,

Topical anesthetics are effective only on surface tissues (2 to 3 mm).

Surface anesthesia does allow for atraumatic needle penetration of the mucous membrane.

- Benzocaine

  Ester local anesthetic drug
- Poor water solubility
- Not suitable for injection
- Available as aersol,gel,ointment
- Dose 140mg/ml

Lidocaine

- Lidocaine base 5%
- Lidocaine water soluble 2% concentration
- maximum recommended dose for topical application is 200 mg.
- Lidocaine base is available as an aerosol spray,ointment,solutions.